THE LOCALIZED CHERN CLASSES ARE COMPATIBLE WITH BASE CHANGE OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FB3 (sources/stacks-chow/pinned-original-source.tex line 10824).

Allowed standard prerequisites (editorial).

- Construction of the localized Chern classes from the blowing up and the complex Q. Same chapter, Lemma labelled lemma-localized-chern-pre and Lemma labelled lemma-independent-loc-chern, cited at proof lines 10868-10873. Identifies the classes being compared with P'_p(Q) and P'_p(Q').
- Base change of the blowing up complex and the description of Q as a derived twist. More on Flatness, Lemma labelled flat-lemma-complex-and-divisor-blowup-pre and Lemma labelled flat-lemma-complex-and-divisor-blowup-base-change, cited at proof lines 10843-10865. Provides the commutative square with Q' = (q o g)^*Q that realises the base change of the construction.
- Base change of the localized class itself and independence of the chosen closed subscheme T'. Same chapter, Lemma labelled lemma-base-change-localized-chern-pre, cited at proof lines 10887-10893, and Lemma labelled lemma-localized-chern-pre-independent, cited at proof lines 10882-10886 and 10897-10900. Compares the restriction of P'_p(Q) with the class built from q^*Q and shows P'_p(Q') = P'_p(q^*Q).
- Restriction of bivariant classes. Same chapter, Remark labelled remark-restriction-bivariant, invoked in the statement. Fixes the meaning of the comparison asserted by the lemma.

Source reference. Original reference 0GUJ: sources/stacks-chow/pinned-original-source.tex, source lines 10678–10678. Same-chapter pointer into the author environment situation at source line 10678; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original definition 0FB5: sources/stacks-chow/pinned-original-source.tex, source lines 10787–10787. Same-chapter pointer into the author environment definition at source line 10787; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FB2: sources/stacks-chow/pinned-original-source.tex, source lines 10701–10701. Same-chapter pointer into the author environment lemma at source line 10701; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Same-chapter pointer into the author environment remark at source line 6184; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0GTY: sources/stacks-chow/tags.txt, source lines 20983–20983. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10848 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F8V: sources/stacks-chow/tags.txt, source lines 19055–19055. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10861 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma at source line 10204; invoked at proof line 10877 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAU: sources/stacks-chow/pinned-original-source.tex, source lines 10301–10301. Same-chapter pointer into the author environment lemma at source line 10301; invoked at proof line 10886 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0GUI: sources/stacks-chow/pinned-original-source.tex, source lines 10276–10276. Same-chapter pointer into the author environment lemma at source line 10276; invoked at proof line 10889 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0GUJ let $f : X' \to X$ be a morphism of schemes which is locally of finite type. Denote $E' = f^*E$ and $Z' = f^{-1}(Z)$. Then the bivariant class of Definition 0FB5*

$$P_p(Z' \to X', E') \in A^p(Z' \to X'), \quad \text{resp.} \quad c_p(Z' \to X', E') \in A^p(Z' \to X')$$

constructed as in Lemma 0FB2 using X', Z', E' is the restriction (Remark 0F9Z) of the bivariant class $P_p(Z \to X, E) \in A^p(Z \to X)$, resp. $c_p(Z \to X, E) \in A^p(Z \to X)$.

Proof. Denote $p : \mathbf{P}^1_X \to X$ and $p' : \mathbf{P}^1_{X'} \to X'$ the structure morphisms. Recall that $b : W \to \mathbf{P}^1_X$ and $b' : W' \to \mathbf{P}^1_{X'}$ are the morphism constructed from the triples $(\mathbf{P}^1_X, (\mathbf{P}^1_X)\infty, p^*E)$ and $(\mathbf{P}^1_{X'}, (\mathbf{P}^1_{X'})\infty, (p')^*E')$ in More on Flatness, Lemma 0GTY. Furthermore $Q = L\eta_{\mathcal{I}_\infty} p^*E$ and $Q = L\eta_{\mathcal{I}'_\infty}(p')^*E'$ where $\mathcal{I}_\infty \subset \mathcal{O}_W$ is the ideal sheaf of W_∞ and $\mathcal{I}'_\infty \subset \mathcal{O}_{W'}$ is the ideal sheaf of W'_∞. Next, $h : \mathbf{P}^1_{X'} \to \mathbf{P}^1_X$

is a morphism of schemes such that the pullback of the effective Cartier divisor $(\mathbf{P}^1_X)_\infty$ is the effective Cartier divisor $(\mathbf{P}^1_{X'})_\infty$ and such that $h^*p^*E = (p')^*E'$. By More on Flatness, Lemma 0F8V we obtain a commutative diagram

$$\begin{array}{ccccc} W' & \xrightarrow{g} & \mathbf{P}^1_{X'} \times_{\mathbf{P}^1_X} W & \xrightarrow{q} & W \\ & \searrow^{b'} & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle b} \\ & & \mathbf{P}^1_{X'} & \longrightarrow & \mathbf{P}^1_X \end{array}$$

such that W' is the "strict transform" of $\mathbf{P}^1_{X'}$ with respect to b and such that $Q' = (q \circ g)^*Q$. Now recall that $P_p(Z \to X, E) = P'_p(Q)$, resp. $c_p(Z \to X, E) = c'_p(Q)$ where $P'_p(Q)$, resp. $c'_p(Q)$ are constructed in Lemma 0F9K using b, Q, T' where T' is a closed subscheme $T' \subset W_\infty$ with the following two properties: (a) T' contains all points of W_∞ lying over $X \setminus Z$, and (b) $Q|_{T'}$ is zero, resp. isomorphic to a finite locally free module of rank $< p$ placed in degree 0. In the construction of Lemma 0F9K we chose a particular closed subscheme T' with properties (a) and (b) but the precise choice of T' is immaterial, see Lemma 0FAU.

Next, by Lemma 0GUI the restriction of the bivariant class $P_p(Z \to X, E) = P'_p(Q)$, resp. $c_p(Z \to X, E) = c_p(Q')$ to X' corresponds to the class $P'_p(q^*Q)$, resp. $c'_p(q^*Q)$ constructed as in Lemma 0F9K using $r : \mathbf{P}^1_{X'} \times_{\mathbf{P}^1_X} W \to \mathbf{P}^1_{X'}$, the complex q^*Q, and the inverse image $q^{-1}(T')$.

Now by the second statement of Lemma 0FAU we have $P'_p(Q') = P'_p(q^*Q)$, resp. $c'_p(q^*Q) = c'_p(Q')$. Since $P_p(Z' \to X', E') = P'_p(Q')$, resp. $c_p(Z' \to X', E') = c'_p(Q')$ we conclude that the lemma is true. □